



FLEET & SHORE PERSONNEL TRAINING

Bridge Resource Management Training

Why most collisions and groundings trace back to team failures, not equipment — and the practical BRM skills that close that gap on the bridge.

STCW Table A-II/1

IMO BRM Model Course 1.22

Company SMS

What We Will Cover

01

Understanding BRM

Human factors as the dominant cause of collisions and groundings

02

Bridge Team Roles & Responsibilities

Who does what, and how manning should scale with task complexity

03

Situational Awareness

Building and keeping a shared, accurate picture of the ship's situation

04

Closed-Loop Communication

Orders, read-backs and the formal channels the bridge relies on

05

Passage Planning as a BRM Tool

Turning the plan into the team's shared reference for the whole voyage

06

Challenge & Response

Raising a concern, escalating it, and responding well when challenged

07

Fatigue, Workload & Learning From Failure

Managing human limits, and lessons from real BRM failures



SECTION 1

Understanding BRM

Why the great majority of collisions and groundings trace back to human and team failures, not equipment failures.

01

Human Error as the Dominant Cause

Investigations consistently point to how the bridge team worked together, not just what any one person did wrong.

- Studies of major collisions and groundings attribute somewhere between 75 and 96 percent of casualties to human and organisational factors rather than equipment failure.
- Individually competent officers can still produce an unsafe bridge if information is not shared, cross-checked and acted on as a team.
- Bridge Resource Management is the set of skills that makes full use of all available resources — people, information and equipment — to ensure a safe operation.
- The concept was adapted from Crew Resource Management, developed in commercial aviation after a series of accidents traced to poor team coordination, not technical failure.
- STCW Table A-II/1 and A-II/2 require BRM/BTM as a mandatory competence for all deck officers, not an optional soft skill.



Not Just the Master

BRM shifts responsibility for safety from one individual's vigilance to a whole team's shared situational picture — every member has a duty to speak up.

CORE SKILLS

The Four Pillars of Bridge Resource Management

Each pillar reinforces the others — weak communication undermines situational awareness, and vice versa.



Situational Awareness

Building and maintaining an accurate, shared picture of the ship's situation at all times.



Communication

Clear, closed-loop exchange of information between every member of the bridge team.



Workload Management

Distributing tasks so no one person becomes overloaded or a single point of failure.



Leadership & Assertiveness

Directing the team clearly while remaining genuinely open to being challenged by any member.

WHERE IT CAME FROM

The Origins of BRM

Maritime BRM training borrowed directly from a hard-won lesson in another industry.

1979

United Airlines Flight 173 prompted aviation to formalise Crew Resource Management after a fatal, preventable crash

1990s

Maritime BRM programmes adapted CRM principles for the bridge after a string of collisions and groundings with similar root causes

STCW

BRM/BTM became a mandatory STCW competence, examined and assessed like any other navigational skill



SECTION 2

Bridge Team Roles & Responsibilities

02

Who does what on a well-run bridge, and how manning should scale as the task gets harder.

DEFINING ROLES

Who Does What on a Well-Run Bridge

Clear roles, agreed before a critical phase begins, are one of the simplest and most effective BRM tools available.

- The master retains overall command and responsibility at all times, even when another officer is conning the ship or a pilot is embarked.
- The Officer of the Watch (OOW) is responsible for the safe navigation of the ship during the watch, following the master's standing and night orders.
- A dedicated lookout, separate from the OOW, must be maintained at all times by sight and hearing, as required by COLREG Rule 5.
- Additional bridge team members — a second officer, helmsman or pilot — are added as traffic density, visibility or manoeuvring complexity increase.
- Every role should be explicitly assigned and understood before a critical phase of the passage, not assumed or left to default to whoever is present.



Bridge Manning

Manning levels should scale up in restricted visibility, heavy traffic or during pilotage, not stay fixed regardless of task complexity.

Assigning and Respecting Roles

Ambiguity about who is doing what is one of the fastest ways a bridge team's safety margin quietly disappears.



Do

- Brief who holds the conn, who navigates and who keeps lookout before the watch begins
- Change bridge manning proactively as conditions or workload increase
- Confirm handover of the conn verbally and explicitly, never by assumption
- Keep the lookout role free of other competing tasks



Don't

- Don't let the OOW work as sole lookout in restricted visibility or heavy traffic
- Don't assume everyone understands their role without a stated brief
- Don't let rank alone decide who checks whom — every team member must be able to challenge
- Don't leave the bridge unmanned or under-resourced during a critical manoeuvre

MANNING LEVELS

Matching Team Size to Task

Bridge manning is a deliberate decision, reviewed as the situation changes, not a fixed roster.

Situation	Minimum Team	Why
Open water, good visibility, light traffic	OOW + lookout	Standard watchkeeping provides an adequate safety margin
Restricted visibility or heavy traffic	OOW + dedicated lookout + additional officer	Extra capacity for plotting, radar monitoring and communication
Pilotage, berthing or restricted waters	Master + pilot + OOW + helmsman + lookout	Full team needed for close manoeuvring and cross-checking



SECTION 3

Situational Awareness

Building and keeping a shared, accurate picture of the ship's situation — and noticing early when that picture starts to slip.

03

What Situational Awareness Actually Means

It is a skill that is actively built and maintained, not a state that simply happens by being on watch.

- Situational awareness is the accurate perception of what is happening around the ship, what it means, and what is likely to happen next.
- It is built from multiple sources — visual lookout, radar/ARPA, ECDIS, AIS, VHF traffic and the passage plan — cross-checked against each other.
- Loss of situational awareness rarely happens all at once; it typically erodes gradually as attention narrows onto a single task or display.
- A shared mental model means every bridge team member holds broadly the same picture of the ship's position, intentions and surrounding traffic.
- Regular, brief verbal updates — 'position check', 'traffic on the port bow' — keep the whole team's picture current, not just the conning officer's.



Tunnel Vision

Fixating on one instrument or task — a single radar target, an alarm, a phone call — is one of the most common ways situational awareness is lost.

Three Levels, Applied to the Bridge

A widely used model for situational awareness maps directly onto everyday watchkeeping.



Perception

Noticing the relevant elements — other vessels, aids to navigation, depth, weather — in the environment around the ship.



Comprehension

Understanding what those elements mean together: is that vessel on a closing course, is the tide about to turn?



Projection

Anticipating how the situation will develop — where will that target be in ten minutes if nothing changes?



Loss of SA — Warning Signs

Confusion, no one asking questions, fixation on one task, or reliance on a single source of information.

Keeping the Whole Team Aware

Situational awareness is a team property, not an individual one — it has to be actively shared to exist at all.



Do

- Cross-check instruments against each other and against the view out of the window
- Verbalise significant changes — course alterations, new contacts, alarms — out loud
- Rotate or share tasks so no one person is overloaded on a single display
- Step back periodically and reassess the overall picture, not just the last alarm



Don't

- Don't rely on a single sensor or source of information for a critical decision
- Don't let radio or admin tasks pull attention away from lookout in close-quarters situations
- Don't assume silence from the team means everyone shares your picture
- Don't continue a plan without re-checking it once new information contradicts it



SECTION 4

Closed-Loop Communication

Orders, read-backs and the formal channels the bridge relies on to keep information flowing accurately.

04

Common Communication Breakdowns on the Bridge

Most communication failures on the bridge are quiet ones — an order given but never truly confirmed.

- An order given but not acknowledged leaves the giver unsure whether it was heard, understood, or is being carried out at all.
- Ambiguous phrasing — 'slow down a bit', 'watch that ship' — leaves room for two people to interpret the same words differently.
- Hierarchical gradient, the reluctance of a junior officer to question a senior's order, silences exactly the input that might prevent an error.
- Non-native English speakers on a multinational bridge benefit especially from standard phrases and explicit read-back, not idiom or slang.
- Distractions — personal conversation, phone use, unrelated radio traffic — interrupt the closed-loop exchange at the worst possible moment.



IMO Standard Phrases

The IMO Standard Marine Communication Phrases exist precisely to remove ambiguity — use them for helm orders, engine orders and traffic communication.

Every Order, Every Time

Closing the loop takes a few extra seconds and removes almost all doubt about whether an order landed correctly.



Do

- State the order clearly, using standard phrases where they exist
- Require the receiver to read the order back before acting on it
- Confirm out loud once the order has been carried out
- Ask for clarification immediately if any part of an order is unclear



Don't

- Don't accept a mumbled or partial acknowledgement as confirmation
- Don't give two orders at once, or a new order before the first is complete
- Don't assume understanding just because no question was asked
- Don't let rank or urgency skip the read-back step

Formal Channels the Bridge Relies On

Structured exchanges catch what casual conversation reliably misses.



Master/Pilot Exchange

A structured briefing before the pilot takes the conn, covering passage plan, manoeuvring data and any ship defects.



Watch Handover

A deliberate, checklist-supported exchange of the navigational picture between outgoing and incoming officers of the watch.



VHF Bridge-to-Bridge

Standard phrases for agreeing passing arrangements with other vessels in sight, especially in narrow channels.



Standing & Night Orders

The master's written instructions defining when the OOW must call the master, kept current and briefed to every officer.



SECTION 5

Passage Planning as a BRM Tool

Turning the plan into the whole team's shared reference for the voyage, not one officer's private working paper.

05

Why the Passage Plan Is a Team Document

A plan that lives only in one officer's head or notebook cannot support the rest of the team's situational awareness.

- A passage plan produced solely by one officer and briefed to no one gives the rest of the team nothing to cross-check against during the voyage.
- The four-stage IMO planning process — appraisal, planning, execution, monitoring — gives BRM a structured framework rather than an informal habit.
- Berth-to-berth planning should include no-go areas, parallel indexing references, contingency anchorages and abort points, not just a track line.
- Briefing the plan to the full bridge team before departure builds the shared mental model that situational awareness depends on throughout the voyage.
- Monitoring is where BRM does its real work: continuously comparing the ship's actual position and progress against the plan, and speaking up on any divergence.



Plan the Deviation, Too

Decide in advance how much cross-track error, however small, triggers a question to the conning officer — don't leave the threshold to individual judgement in the moment.

Appraisal, Planning, Execution, Monitoring

The four stages apply to every voyage, whether an ocean passage or a short coastal hop.

1

Appraisal — gathering all information relevant to the intended voyage, from charts to weather to company requirements

2

Planning — the detailed track, with margins, contingencies and no-go areas, prepared well ahead of departure

3

Execution — carrying out the plan, adjusted as needed for actual conditions on the day

4

Monitoring — continuously checking the ship's actual position and progress against the plan

Using the Plan as a BRM Tool

The plan only functions as a safety net if the whole team treats it as a live reference, not a formality.



Do

- Brief the full plan to the bridge team before departure, not just post it on the chart table
- Agree in advance what cross-track or timing deviation requires the master to be called
- Update the plan and re-brief it when conditions or the situation change materially
- Use parallel indexing and clearing bearings as an independent monitoring cross-check



Don't

- Don't treat the plan as a formality completed only to satisfy an audit or inspection
- Don't deviate from the plan without discussing it with the rest of the bridge team
- Don't rely on a single means of position-fixing to monitor progress against the plan
- Don't let an outdated plan stay in use once the passage has materially changed



SECTION 6

Challenge & Response

Raising a concern, escalating it firmly when needed, and responding well when someone challenges you.

06

Why a Questioning Attitude Is a Skill, Not a Trait

A questioning attitude can be taught, practised and expected — it does not depend on any one person's personality.

- A questioning attitude means actively looking for information that contradicts the current plan or expectation, not passively waiting for something to go wrong.
- Challenge and response is a structured way of raising a concern and requiring the recipient to respond, rather than letting a raised concern go unanswered.
- Steep hierarchical gradients — real or perceived — are consistently identified in accident investigations as a reason valid concerns were never voiced.
- The person who challenges is not accusing the other of error; they are contributing another data point to the team's shared situational picture.
- A culture that welcomes challenge, rather than punishing it, is built deliberately by leaders who visibly respond well when challenged themselves.



Graded Assertiveness

Raise a concern, then escalate firmly if it is not addressed — from a simple question, to a stated concern, to a direct statement that safety requires action now.

The Escalation Ladder, From a Question to Taking Over

Each step is used only if the one before it fails to resolve the concern.



1. Ask

Open with a simple question to prompt the other person to reconsider: 'Are we clear of that vessel?'



2. State a Concern

If the question doesn't resolve it, state the concern directly: 'I'm not comfortable with our CPA to that ship.'



3. State a Solution

Propose the specific action needed: 'I think we should alter course to starboard now.'



4. Take Emergency Action

If there is still no response and danger is immediate, take the necessary action and inform the team.

Responding Well When You're Challenged

How a challenge is received determines whether the next one is ever raised.



Do

- Acknowledge the concern explicitly, even if you believe the situation is under control
- Explain your reasoning briefly so the team's shared picture is restored
- Thank team members, visibly, for raising concerns — even minor or mistaken ones
- Change the plan without hesitation if the challenge turns out to be justified



Don't

- Don't dismiss a concern based on the rank or experience of the person raising it
- Don't let irritation at being questioned discourage the next person from speaking up
- Don't treat a challenge as a personal criticism rather than a safety contribution
- Don't wait until after the fact to explain a decision the team questioned in the moment



SECTION 7

Fatigue, Workload & Learning From Failure

Managing the human limits that make or break BRM in practice, and the lessons real casualties leave behind.

07

How Fatigue Degrades BRM Skills

Every BRM skill covered so far degrades measurably once fatigue sets in.

- Fatigue impairs judgement, reaction time and communication well before a person feels subjectively tired enough to notice the effect.
- STCW rest-hour requirements set minimum rest periods, but compliance on paper does not guarantee the officer is actually well rested.
- Circadian disruption from rotating watches, port calls and cargo operations compounds simple sleep debt, especially on short sea routes.
- A fatigued bridge team is more likely to miss a challenge opportunity, accept an ambiguous order, or fixate on a single task — exactly the failures fatigue makes worse.
- Reporting fatigue honestly, and adjusting watch schedules or manning in response, is a Company SMS responsibility as much as an individual one.



Two-Watch Reality

On a two-watch system, workload management becomes a daily BRM task, not a rare event — plan critical evolutions for when the team is best rested.

Well-Documented BRM Failures in Real Casualties

These widely studied cases are still taught across the industry because the failure was in teamwork, not seamanship.

Casualty	BRM Failure Identified	Lesson for the Bridge Team
Ro-ro ferry capsized, 1987	No positive confirmation before departure; a single unchecked point of failure	Checklist discipline and cross-verification before getting underway
Tanker grounding, 1989	Fatigue and an unsupervised handover to an unqualified watchkeeper	Rest hours, handover discipline and never leaving the bridge unattended
Passenger ship grounding, 2012	Unplanned deviation from the passage plan, unchallenged by the bridge team	Passage plan adherence and a genuine questioning-attitude culture

What the Company Expects on Every Bridge

BRM is not a certificate earned once; it is a daily discipline the whole fleet is expected to practise.

Brief

Every watch and every critical manoeuvre begins with a clear team briefing of roles, plan and known risks

Challenge

Every team member is expected, and protected, to challenge anything that does not match the shared plan

Report

Near-misses and BRM concerns are reported through the SMS without blame, so lessons reach the whole fleet



COURSE COMPLETE

A well-drilled bridge team, not any single officer, is what keeps the ship safe — practise BRM every watch, not just in the classroom.

Company DPA — 24/7 emergency line

Fleet Nautical Training Department — mng@gloryship.com.tr

Fleet Safety Department — mng@gloryship.com.tr